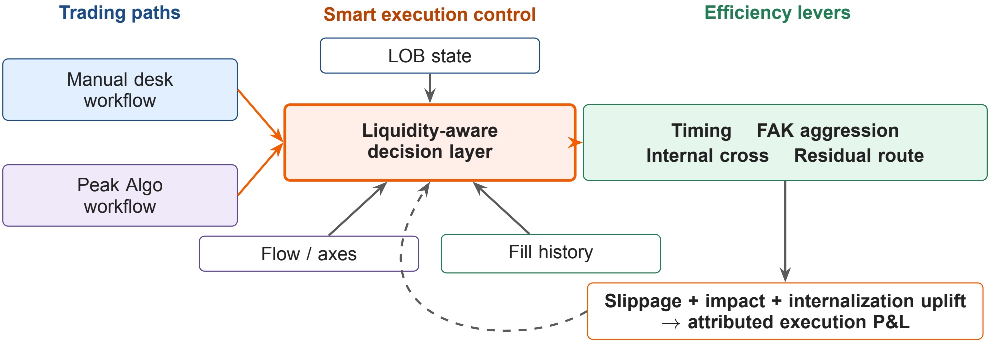
Liquidity-Aware Smart Execution for Desk Efficiency

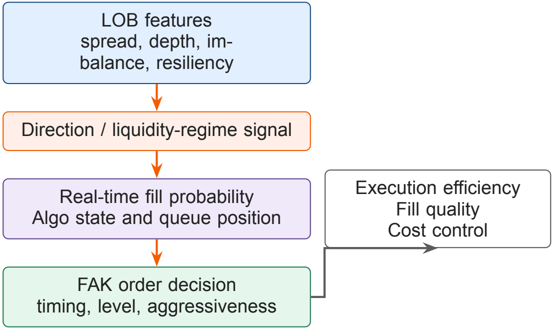
Microstructure Signals Across Manual Trading and Peak Algo

Yida Ding | June 2026

Smart Execution as a Desk Efficiency Layer



FAK Order Execution Efficiency



Case 1: favorable direction

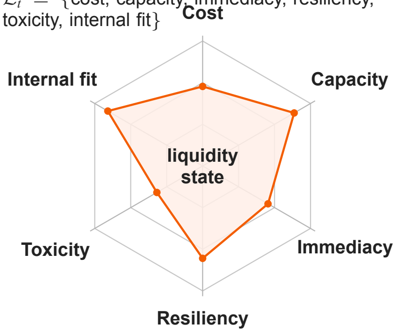
- Direction supports the order side.
- Expected fill probability is higher.
- Execution efficiency: lower slippage and lower market impact.

Case 2: adverse direction

- Direction works against the FAK order.
- Revise fill expectation and aggressiveness.
- Improve success rate and reduce unhedged risk.

Liquidity is the True State Variable

$\mathcal{L}_t = \{\text{cost, capacity, immediacy, resiliency, toxicity, internal fit}\}$



How to read the state

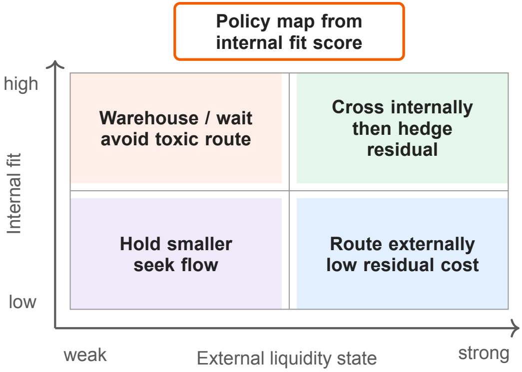
- **High capacity + fast resiliency:** raise FAK confidence.
- **High cost or toxicity:** revise fill expectation and aggression.
- **High internal fit:** cross internally before external routing.
- **Weak state:** delay, slice smaller, or hedge earlier.

Design rule

Price is the observation. Liquidity is the control state.

Source: Lehalle and Laruelle; Cont and de Larrard; Bacry et al.; LOBFrame / LOB benchmark studies.

Internal Flow: Fit Before Route



Fit score inputs

- Orders, RFQ, axes and inventory.
- Side, tenor, size and urgency.
- Liquidity cost, capacity and toxicity.

Decision principle

- Internalize when fit is high.
- Route only the measured residual.
- Prove uplift vs. external cost.

0.01 bp Execution P&L: Attribution Plan

Annual efficiency target

90M

annual target from 0.01 bp saving

- 2025 flow: 45tn.
- Average duration: 2 years.
- $45tn \times 2 \times 0.01bp \approx 90M$.

Replay → Shadow → Pilot → Scale

Where the 0.01 bp comes from

Internal crossing avoid spread + external impact

Liquidity timing depth + resiliency windows

Toxicity control hedge leakage + bad fills

Attribution ledger: slippage, impact, internalization uplift